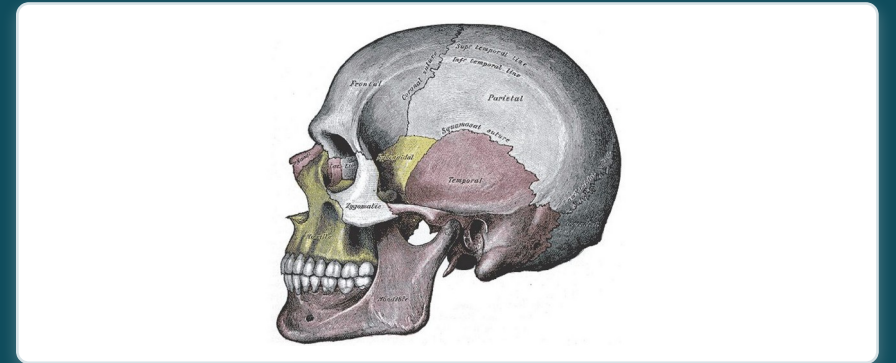


CHAPTER 7

The Axial Skeleton

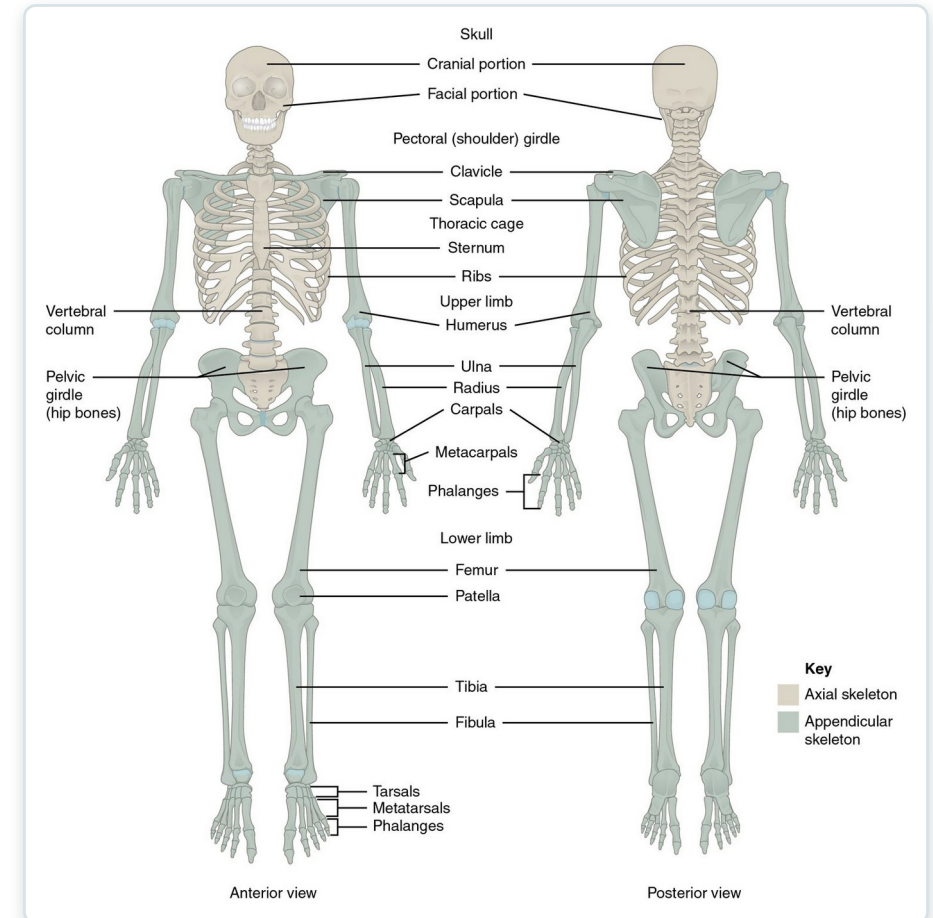
The skull, vertebral column, and thoracic cage.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the functions of the skeletal system and define its two major subdivisions
- 2 Identify the bones and bony structures of the skull, the cranial suture lines, the cranial fossae, and the openings in the skull
- 3 Discuss the vertebral column and regional variations in its bony components and curvatures
- 4 Describe the components of the thoracic cage
- 5 Discuss the embryonic development of the axial skeleton The skeletal system forms the rigid internal framework of the body



The axial skeleton forms the body's central axis.



7.1

SECTION

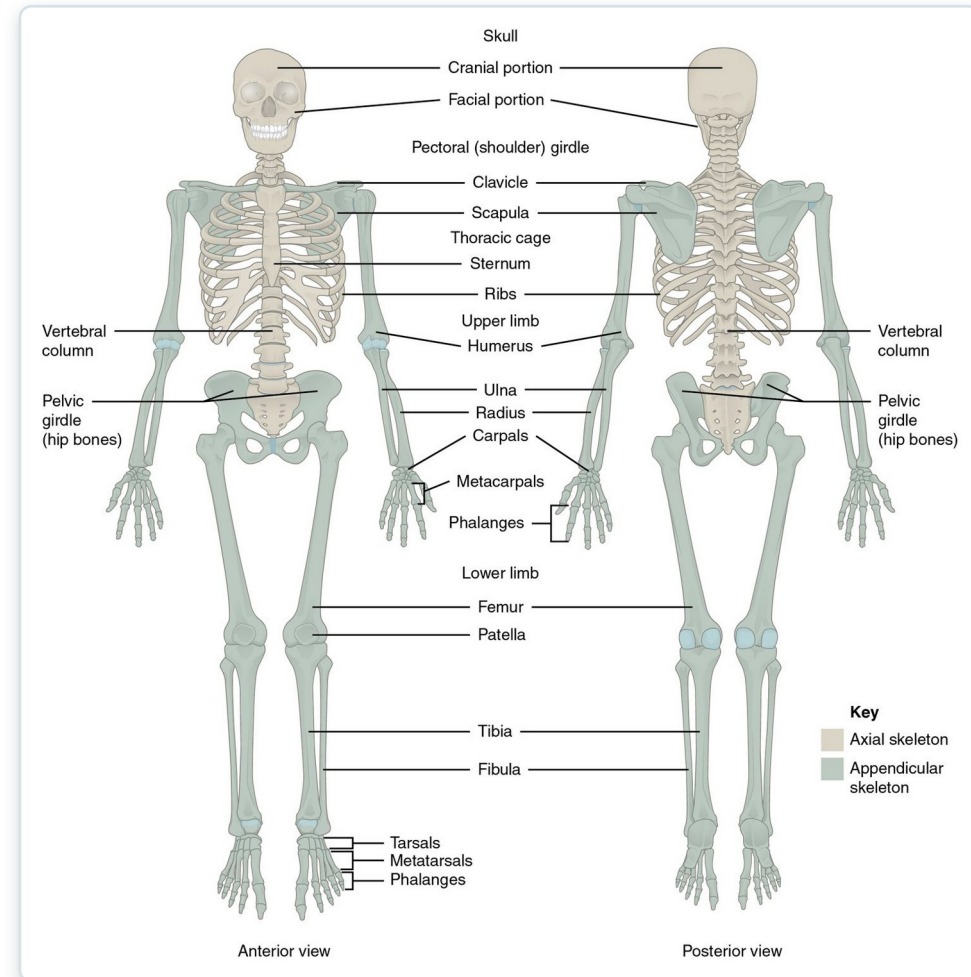
Divisions of the Skeletal System

Dividing the skeleton into the axial core and the appendicular limbs.

SECTION 7.1

Axial vs. Appendicular Skeleton

- 206 bones in two divisions
- Axial: skull, vertebrae, thoracic cage
- Appendicular: limbs and girdles
- The axial forms the central axis



The axial and appendicular skeleton



7.2

SECTION

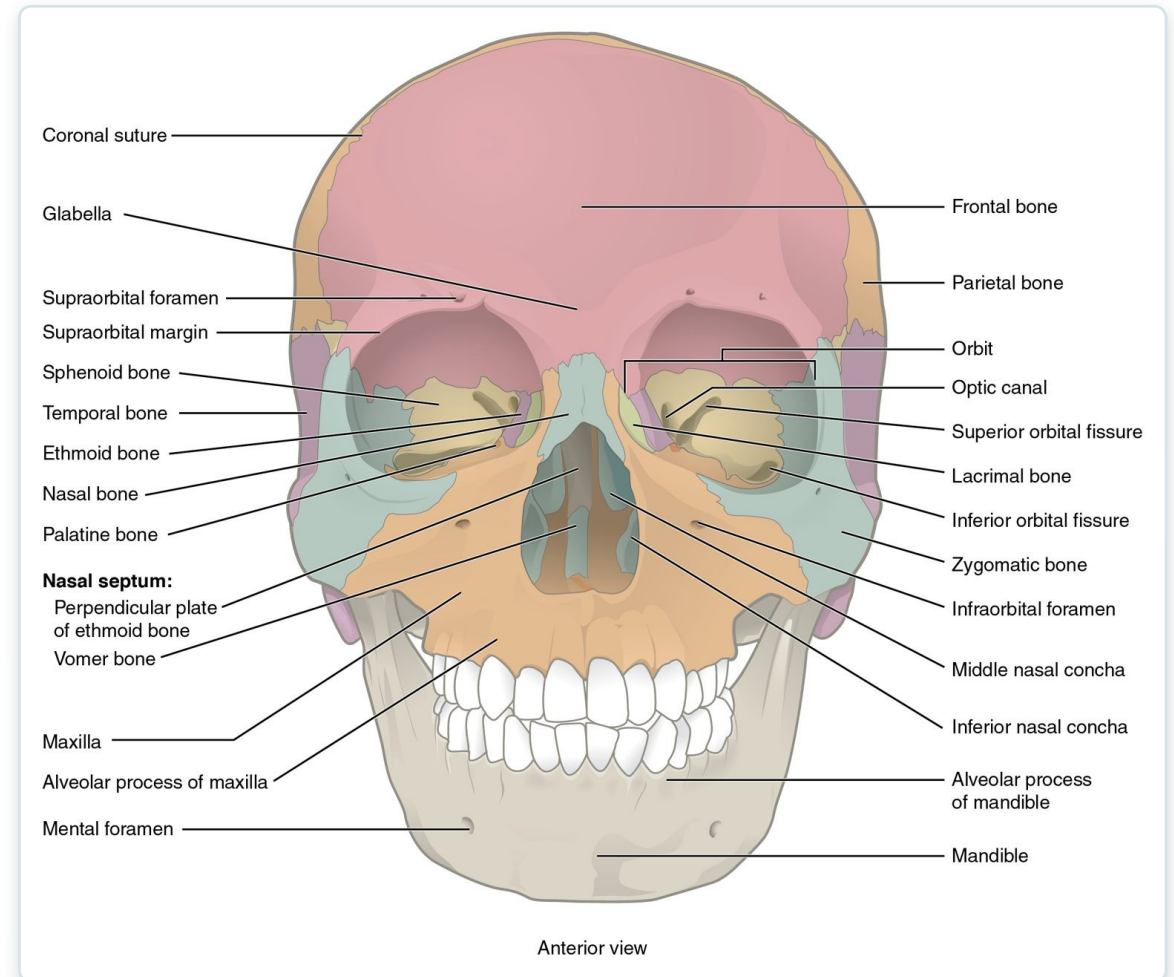
The Skull

The cranial and facial bones that protect the brain and shape the face.

SECTION 7.2

The Skull — Anterior View

- 22 bones: cranial and facial
- Cranial bones enclose the brain
- Facial bones shape the face
- Joined by immovable sutures

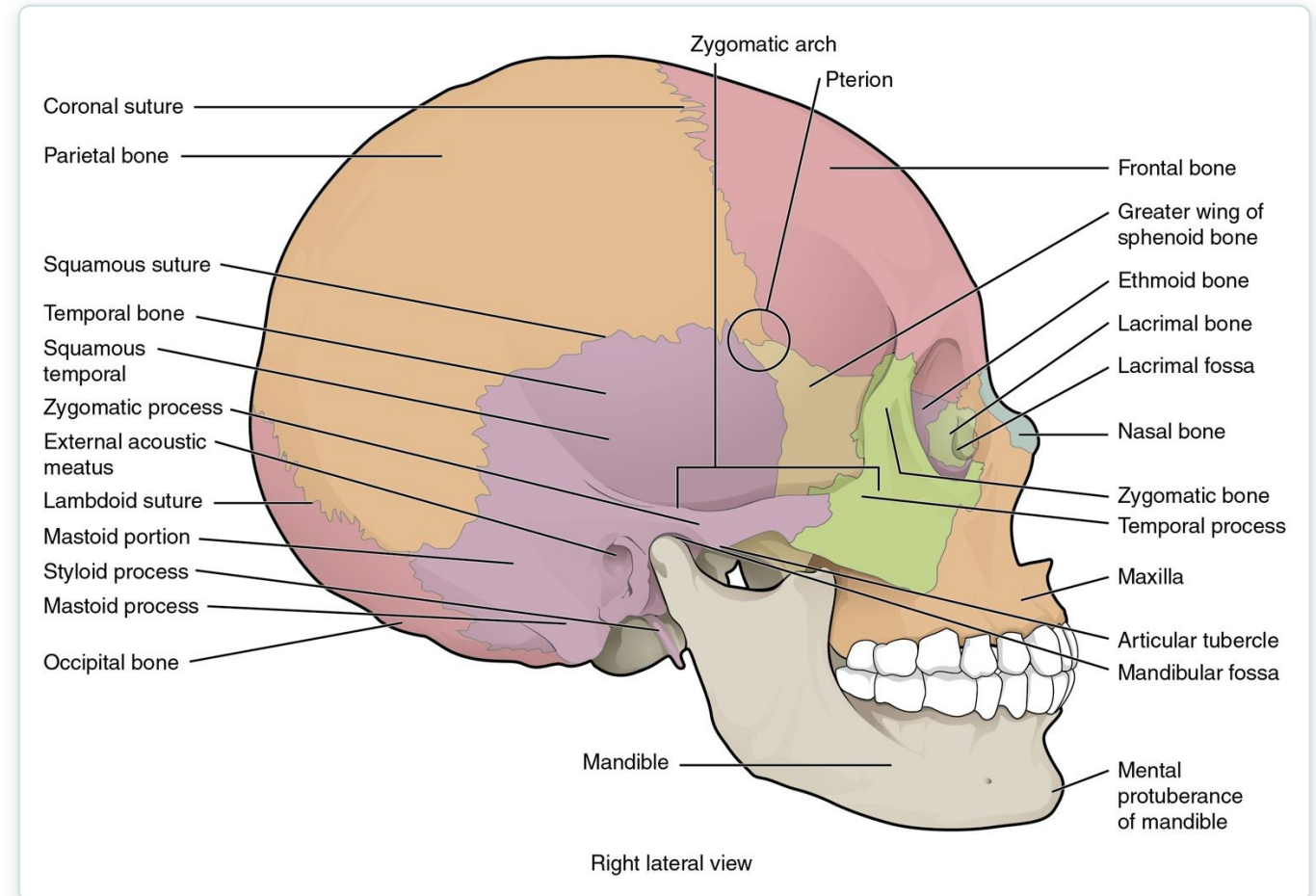


Anterior view of the skull

SECTION 7.2

The Skull — Lateral View

- Frontal, parietal, temporal, occipital
- Sphenoid and ethmoid lie deep within
- Cranial bones meet at sutures
- Cranial fossae cradle the brain

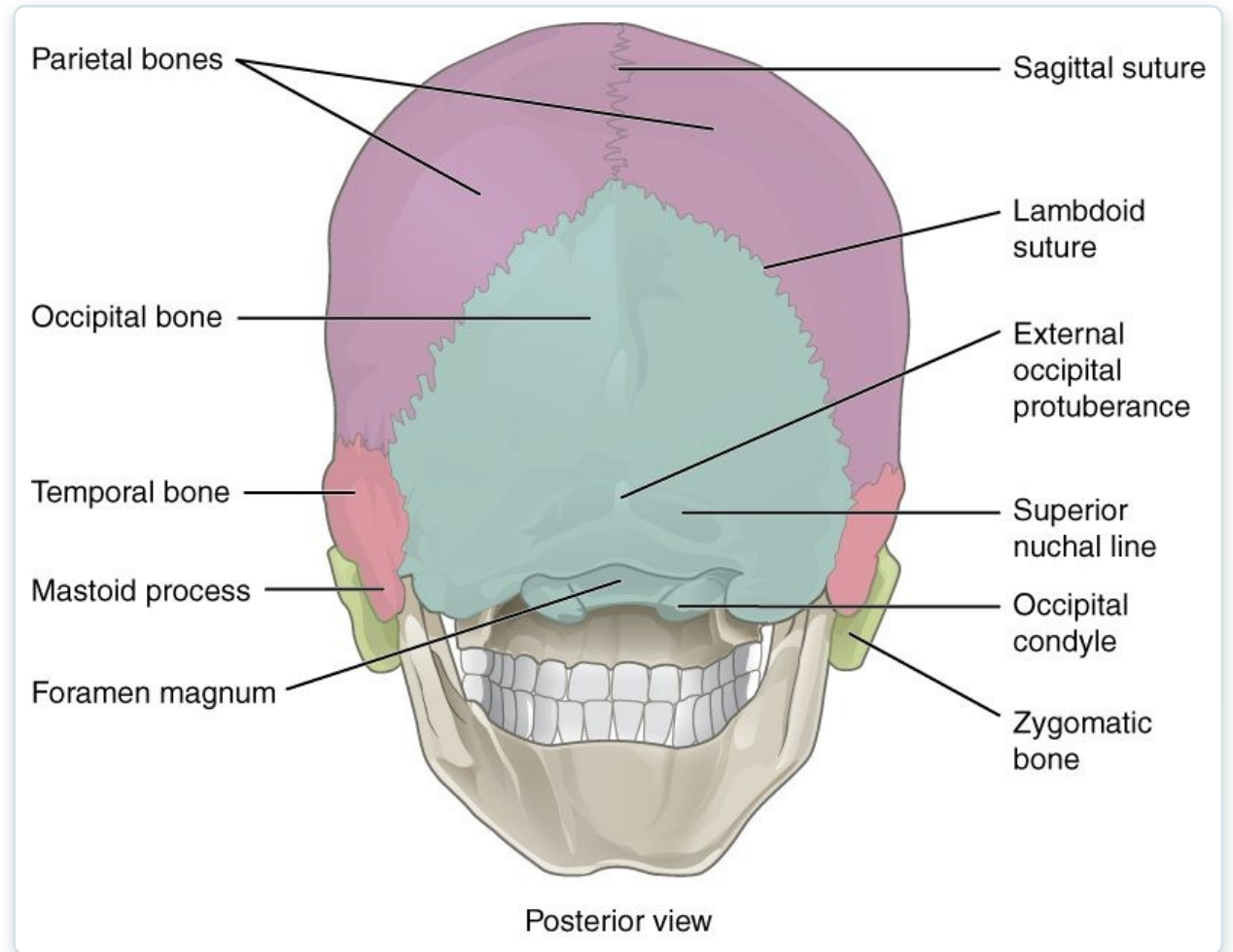


Lateral view of the skull

SECTION 7.2

Sutures of the Skull

- Sutures are immovable fibrous joints
- Coronal, sagittal, and lambdoid
- Lock the cranial bones together
- Foramen magnum at the base

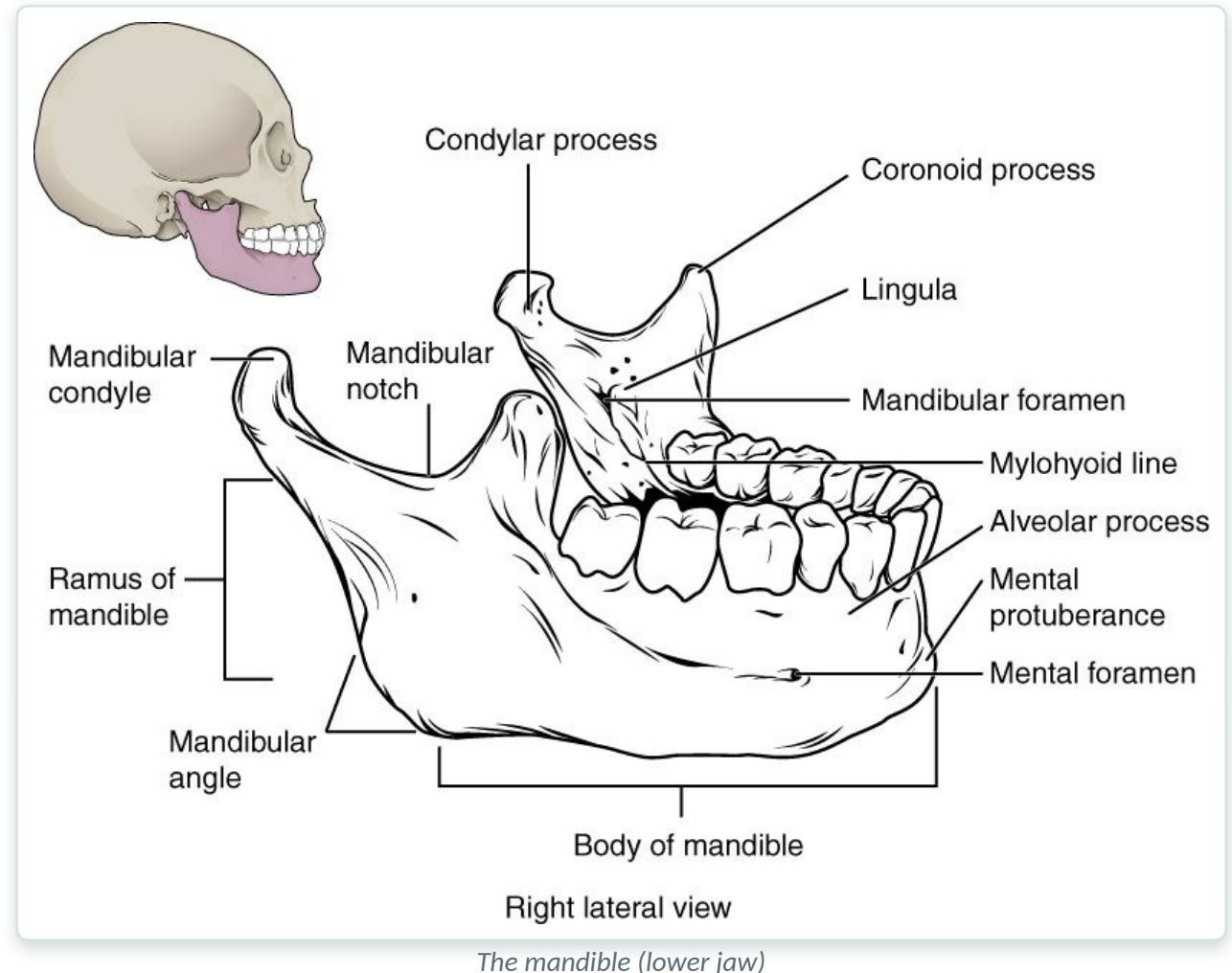


Posterior skull and cranial sutures

SECTION 7.2

The Mandible

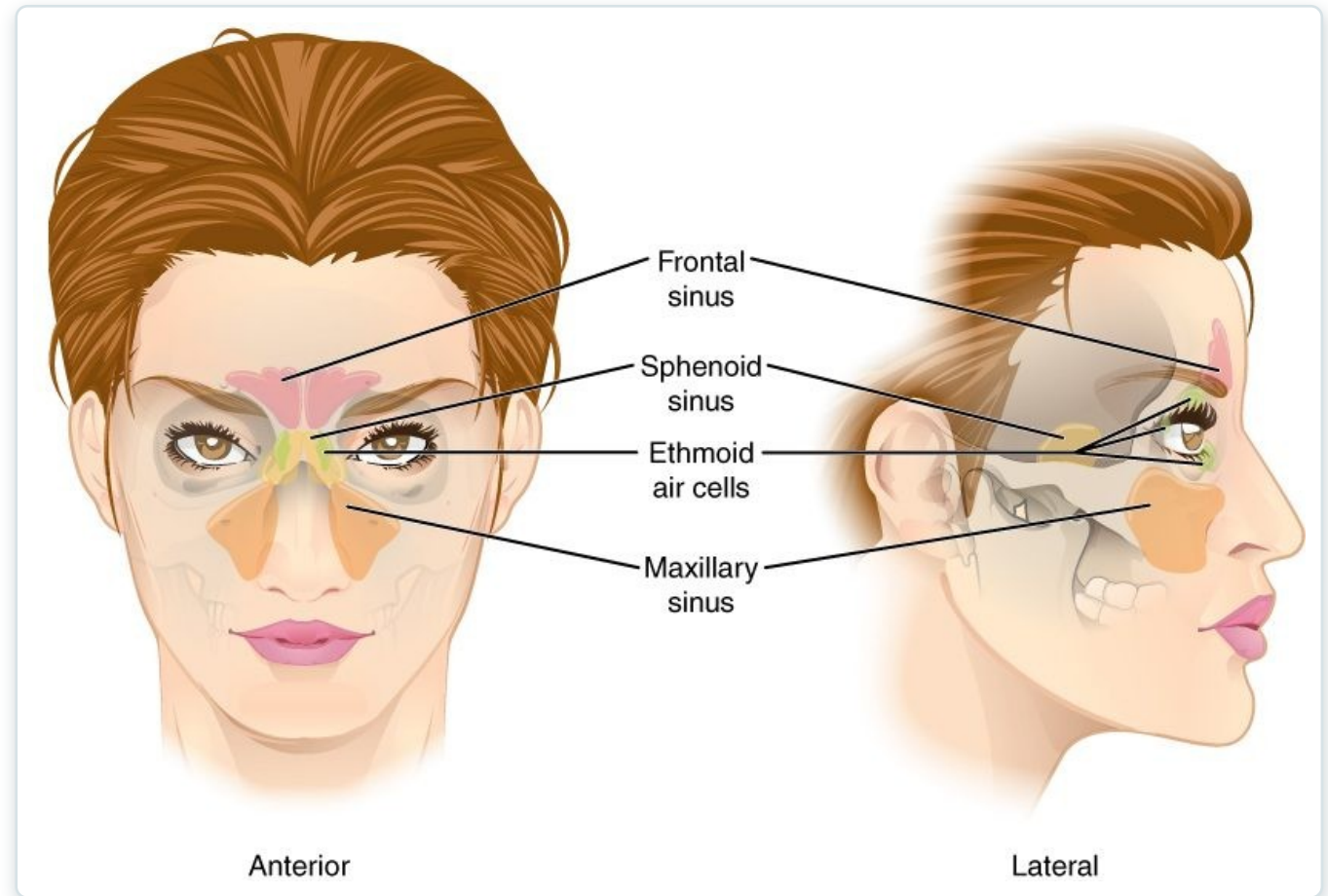
- The only movable skull bone
- Forms the lower jaw
- Holds the lower teeth
- Articulates at the TMJ



SECTION 7.2

The Paranasal Sinuses

- Air-filled spaces in skull bones
- Frontal, ethmoid, sphenoid, maxillary
- Lighten the skull and warm air
- Drain into the nasal cavity



The paranasal sinuses



7.3

SECTION

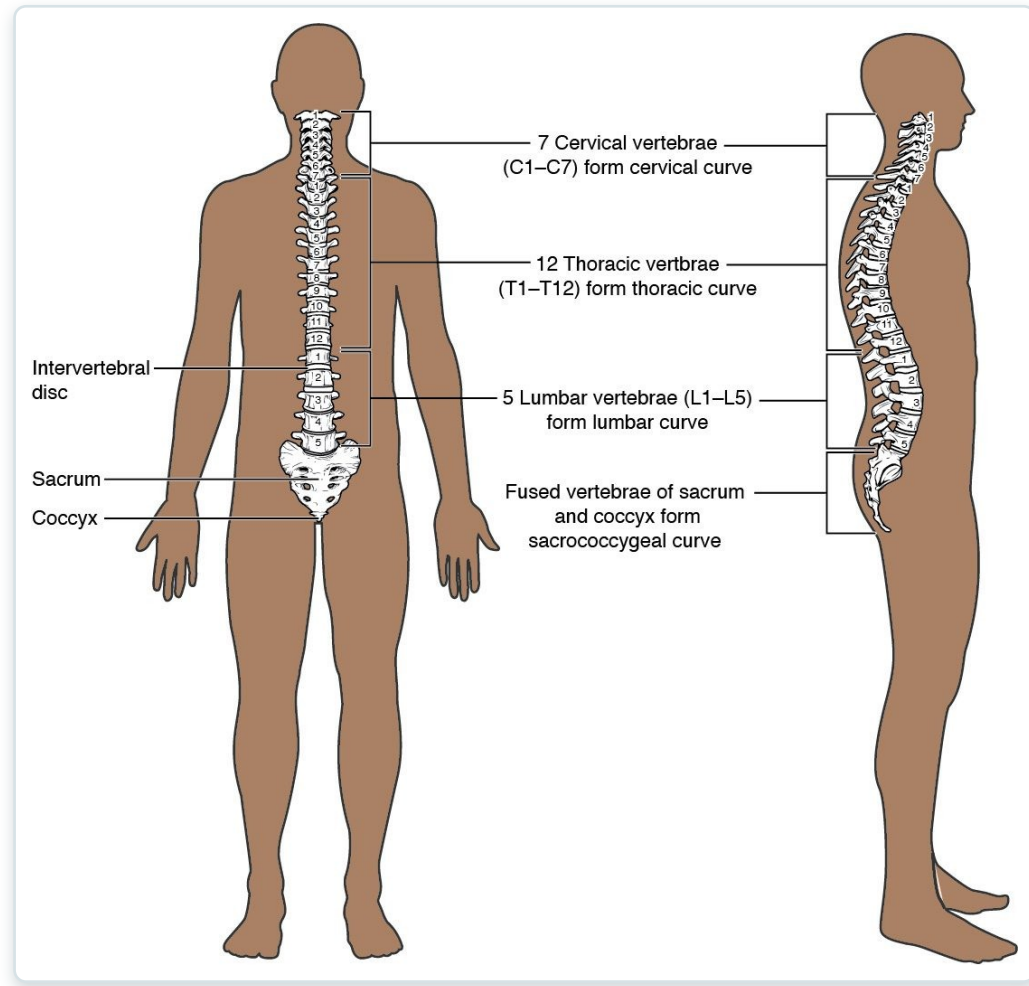
The Vertebral Column

Support, protection of the spinal cord, and flexibility of the spine.

SECTION 7.3

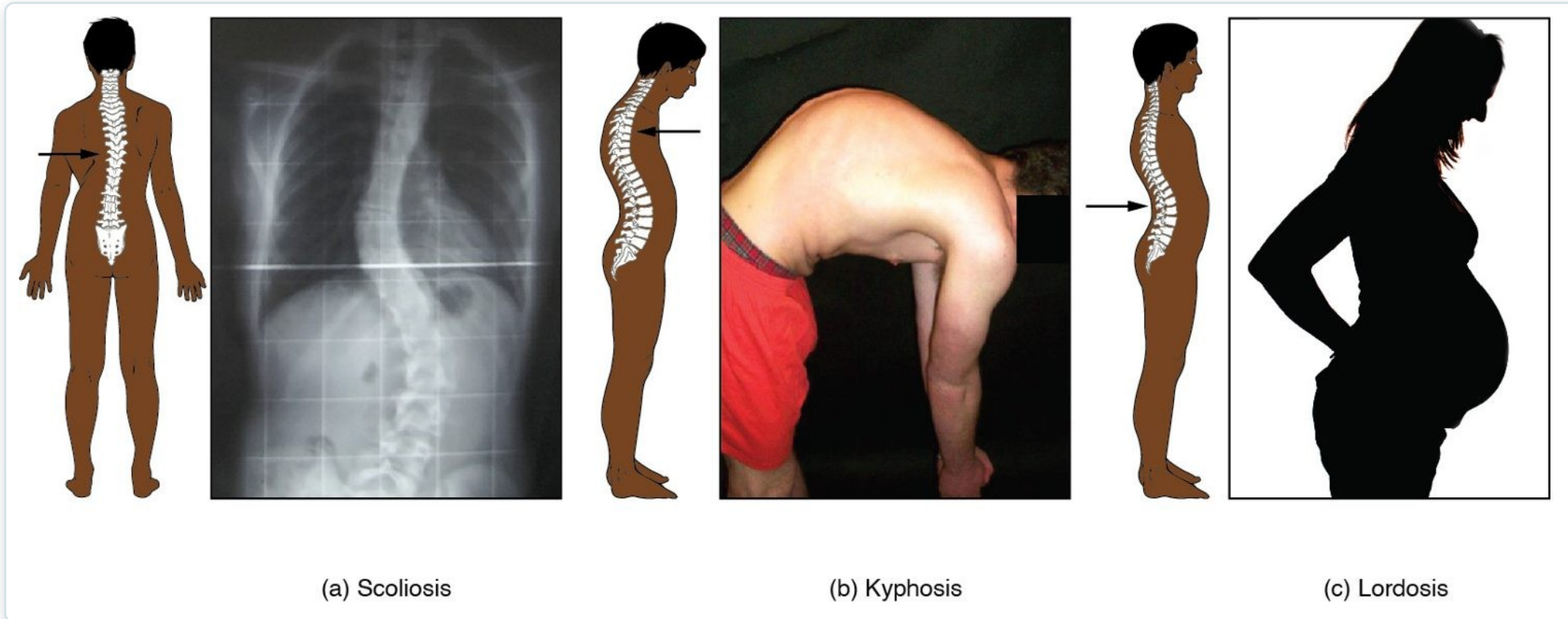
The Vertebral Column

- 26 bones in five regions
- Cervical, thoracic, lumbar, sacrum, coccyx
- Four curvatures balance the body
- Protects the spinal cord



The vertebral column and its curvatures

Abnormal Spinal Curvatures

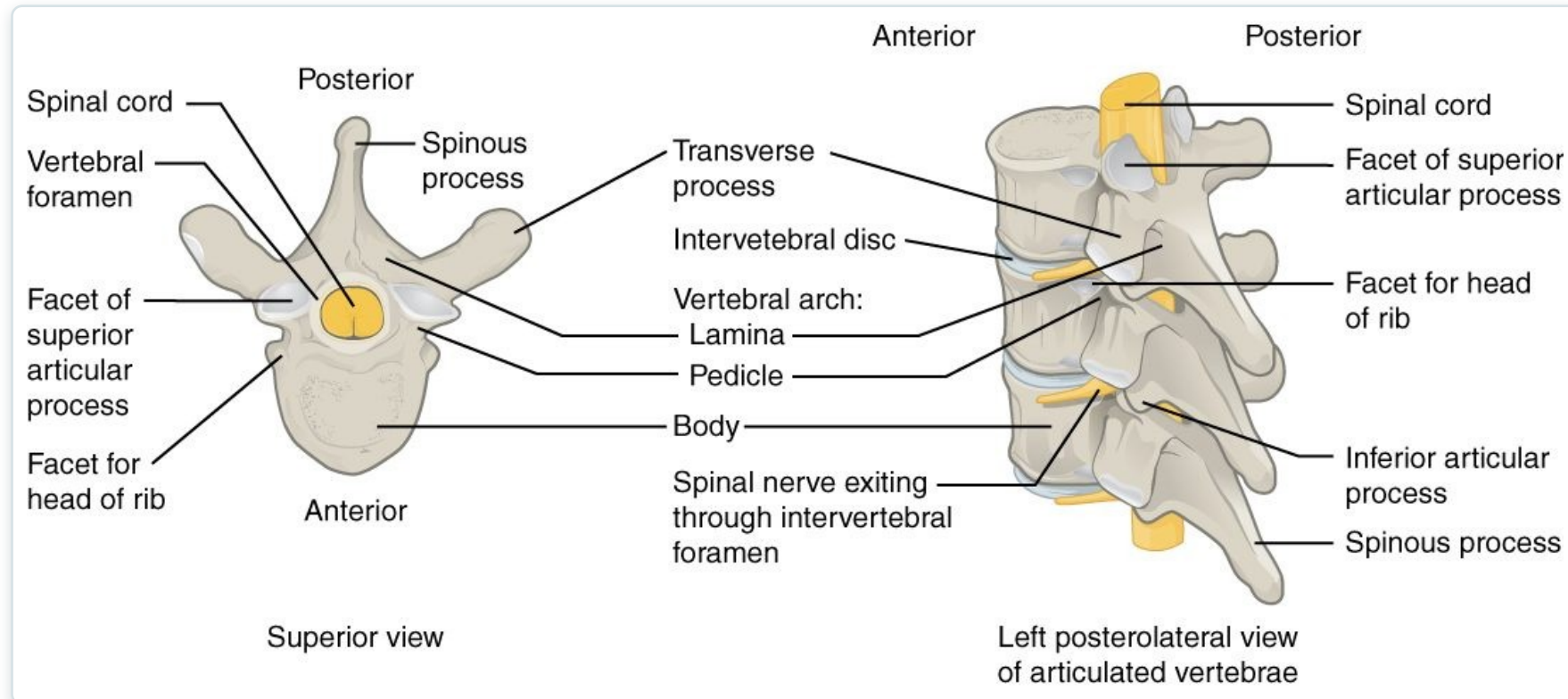


Scoliosis, kyphosis, and lordosis

- Scoliosis: a sideways curvature
- Kyphosis: exaggerated thoracic curve
- Lordosis: exaggerated lumbar curve
- Affect posture and breathing

SECTION 7.3

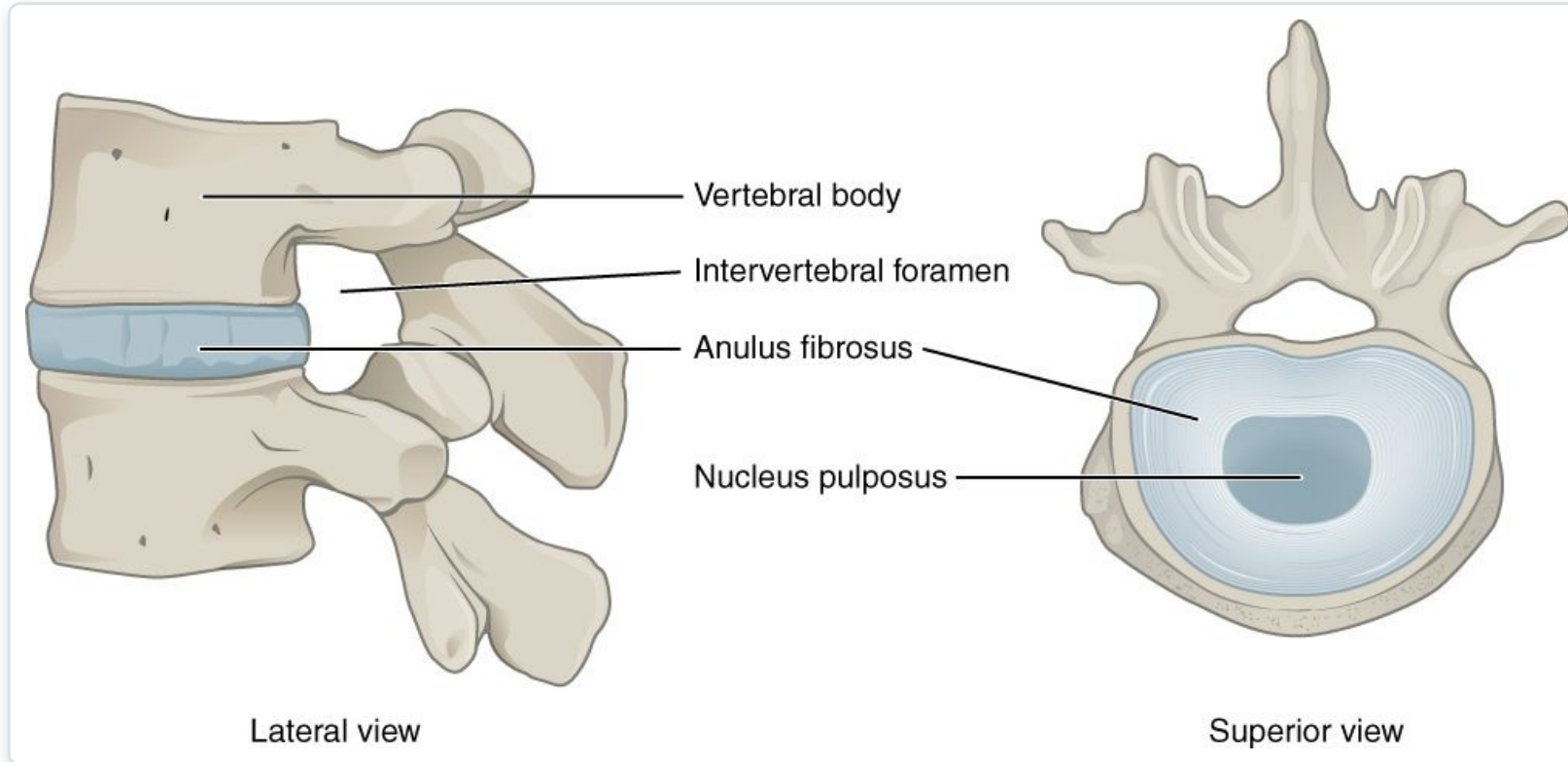
A Typical Vertebra



The structure of a typical vertebra

- Body bears weight in front
- Spinous and transverse processes
- Vertebral arch surrounds the cord
- Stacked foramina form the canal

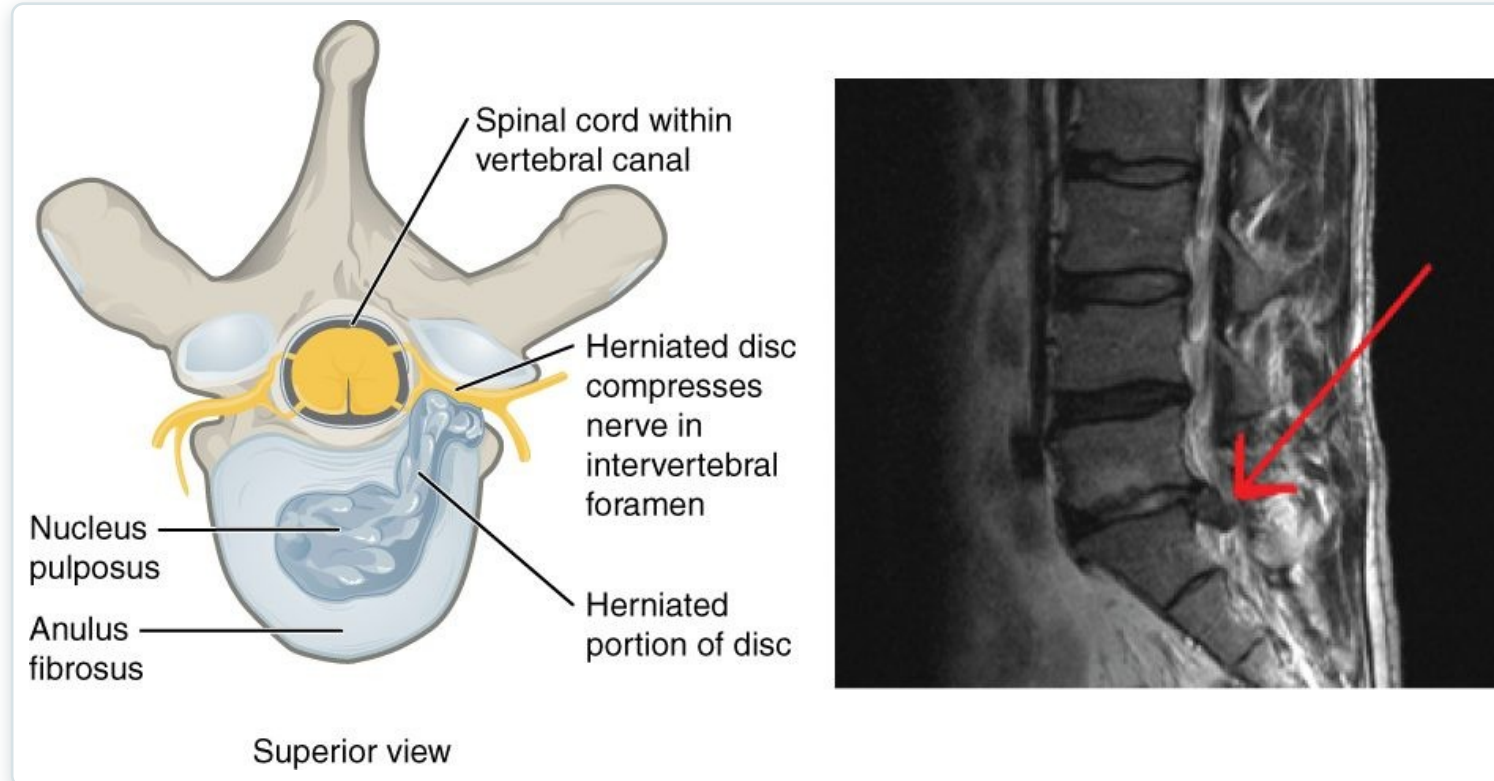
Intervertebral Discs



An intervertebral disc

- Cushions between vertebral bodies
- Tough outer ring (annulus fibrosus)
- Gel-like center (nucleus pulposus)
- Absorb shock and allow movement

Herniated Disc



A herniated intervertebral disc

- Nucleus pushes through the annulus
- Can compress spinal nerves
- Causes pain, numbness, weakness
- Common in the lumbar spine



7.4

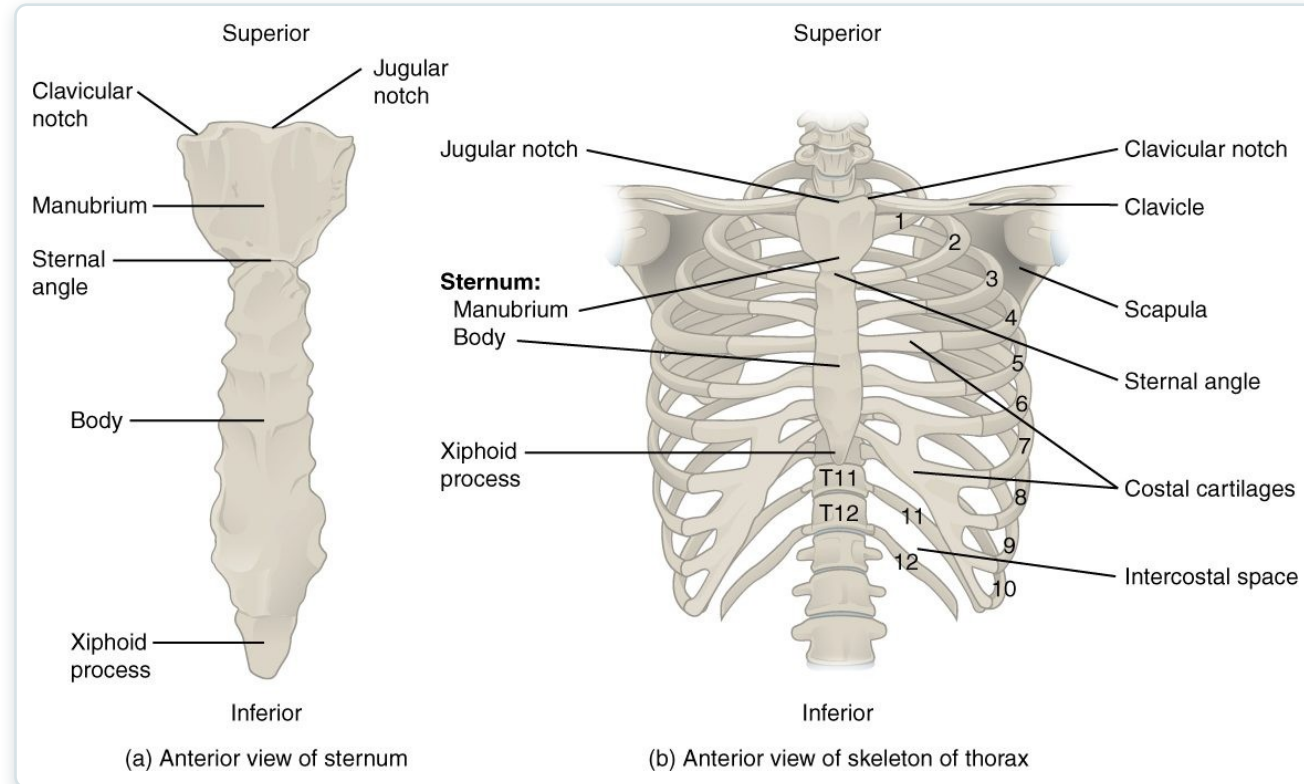
SECTION

The Thoracic Cage

The sternum and ribs that protect the heart and lungs and aid breathing.

SECTION 7.4

The Thoracic Cage



The thoracic cage (sternum and ribs)

- Sternum plus 12 pairs of ribs
- True, false, and floating ribs
- Protects the heart and lungs
- Moves to enable breathing



7.5

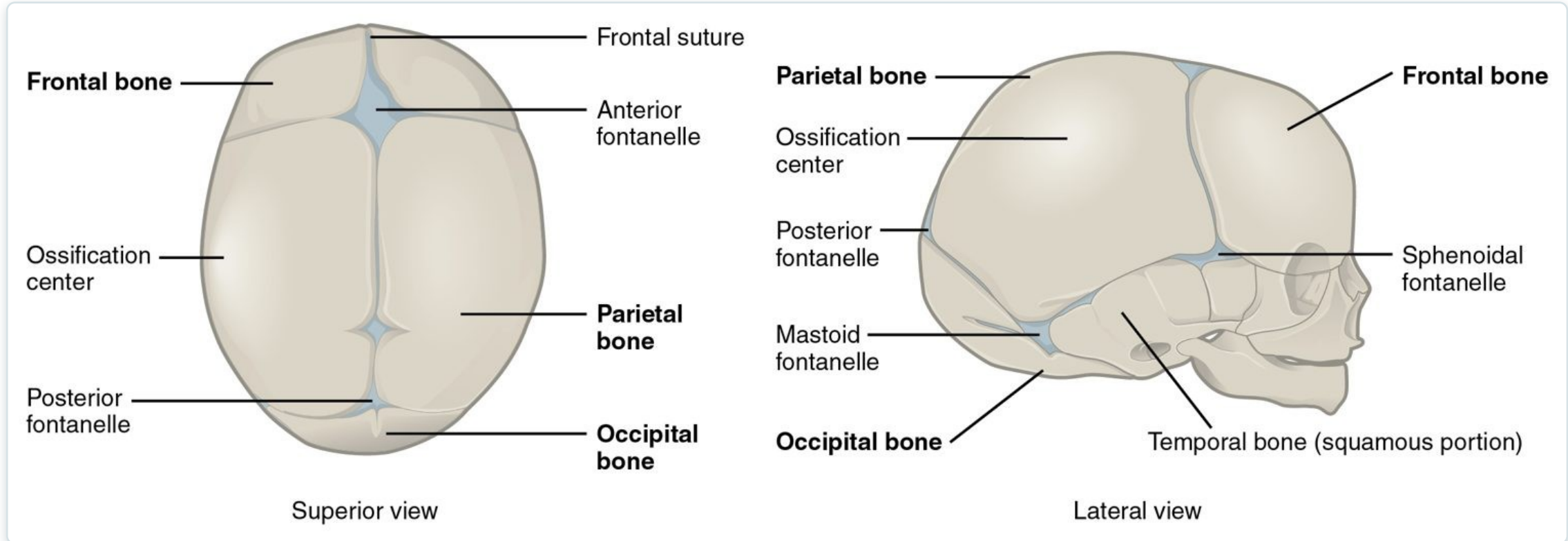
SECTION

Embryonic Development of the Axial Skeleton

How the skull and spine form before birth and in infancy.

SECTION 7.5

Fontanelles & Skull Development



Fontanelles of the newborn skull

- Infant skull bones are not yet fused
- Fontanelles ("soft spots") remain
- Allow birth passage and brain growth
- Close over the first two years

Key Takeaways

- The skeleton divides into the axial (central axis) and appendicular (limbs) divisions.
- The skull's 22 bones protect the brain and form the face, joined by immovable sutures.
- The vertebral column supports the body, protects the spinal cord, and has four curvatures.
- The thoracic cage protects the heart and lungs and assists with breathing.
- The infant skull has fontanelles that allow birth and brain growth.

CLINICAL CONNECTIONS

Why the Axial Skeleton Matters in Practical Nursing

Spine Injuries

Vertebrae protect the spinal cord — spinal precautions prevent paralysis.

Back Pain

Herniated discs and abnormal curvatures are common sources of pain.

Sinusitis

The paranasal sinuses drain into the nose and can become infected.

Infant Assessment

Fontanelles reveal hydration status and intracranial pressure.

Osteoporosis

Thoracic kyphosis often reflects vertebral bone loss in older adults.

CPR Landmarks

The sternum and ribs guide hand placement for chest compressions.